Borrowing Sun

AP6412 Group Assignment Report Draft · Route B · Godot C# · 2026-11-09

Frozen 2D baseline 6175db9 and 3D 05D candidate fb4cbb4. macOS tests and the hosted Windows preflight passed. Manual Windows PC acceptance, 3d-final, group number, member names, and individual reflections are pending.

01 · Game and Original Design

Borrowing Sun is a weather puzzle. Cross the low west bridge, raise the tower with rain, and turn the bridge with wind. Two mirrors relay sunlight from Sun Court (North Court in 3D) to the east receiver, earning a fruit crate. Reuse the rainwater as cargo ballast, ride across the valley, and confirm delivery at Flower Court. Unity L5 inspired the setting and devices. We built and froze the standalone Godot 2D game before porting it to 3D. Game concept

Original assumption	How we test it in both versions
Height and heading give distinct cues	Rain raises and locks the tower at 6 m. Wind changes its dock. Return from Sun Court before turning east.
Players can read the beam path	North lights the tower mirror. East lights the receiver if unobstructed. Both versions calculate mirror hits and occlusion.
Rainwater connects the two parts	After 1.5 seconds of light, load one crate. Of 6 water units, 4 become ballast and 2 stay in the tank. The tower stays locked.
One delivery exercises the loop	Board the car, wait for full ballast and a lit receiver, then press G. Confirm delivery at the Flower Court marker with G to finish at 1/1. Loss of light pauses transfer. A moving car continues.

Controls. Enter or Start begins. Move with A/D in 2D, WASD in 3D, or arrows. Use W/S at the 3D north bridge. Select rain, wind, or sun with 1/2/3, cast with E, depart or confirm delivery with G, and restart at the low bridge with R. Prompts explain invalid actions, and falls return players to safety. The three-minute target awaits first-time playtesting. 2D controls · 3D controls



Frozen 2D baseline · macOS opening screen

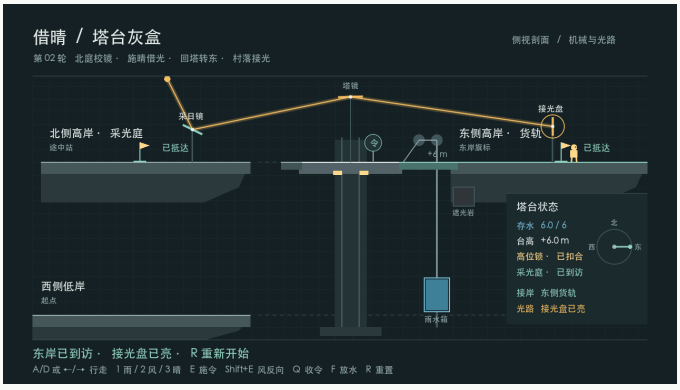


3D 05D candidate · macOS opening screen

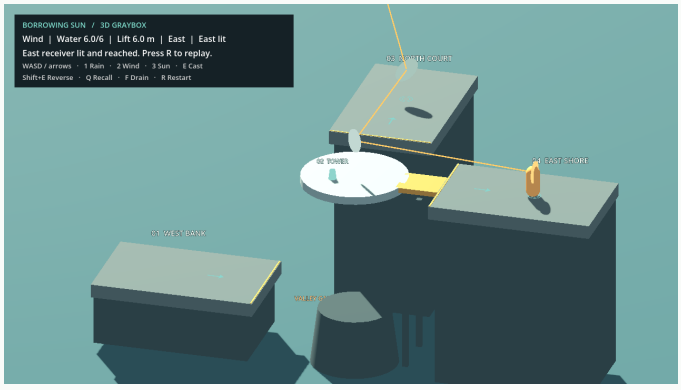
From 2D Prototype to 3D Candidate

02 · Iterations and Design Decisions

Round	Question and decision	Outcome and evidence
00 · Playable 2D	Side view tests the west bridge, rain lift, and wind docking.	Keyboard test exposed a bridge collider left high after R. Same inputs passed after the fix. Failure log · Iteration record
01 · Three shores	Add the east bridge and third dock. Separate bridge sync and visits from the coordinator. Test the RouteProgress finish extension.	Walk west → tower → north → tower → east and retest after reset. Route · SOLID
02 · Double reflection	Calibrate, cast sun, then turn east. Distinguish tower hit from east receiver light.	Verify north, east, and occlusion. 2D optics
03 · Cargo	Add CargoJourney with reused water and separate arrival and confirmation. Integrate tests, CI, and CCR without changing gameplay.	Test loss of light, departure gates, and replay. Cargo · Integration
04 · 2D freeze	Add goal, controls, hints, delivery screen, and replay.	Freeze 2d-baseline at 6175db9. Acceptance
05A · 3D tower	Link tower and route rules. Implement lift, yaw, and north bridge.	Reach Sun Court and repeat after R. 05A
05B · 3D optics	Add east bridge, world-space discs, and occlusion raycasts.	Confirm north and east outcomes, then block and restore the beam. 05B
05C–D · 3D cargo	Reuse cargo and align screens. Fix a stale light prompt and a wrapped character.	Retest loading, refusal, delivery, and replay. Pin fb4cbb4. 05C · 05D



02 · 2D east receiver lit



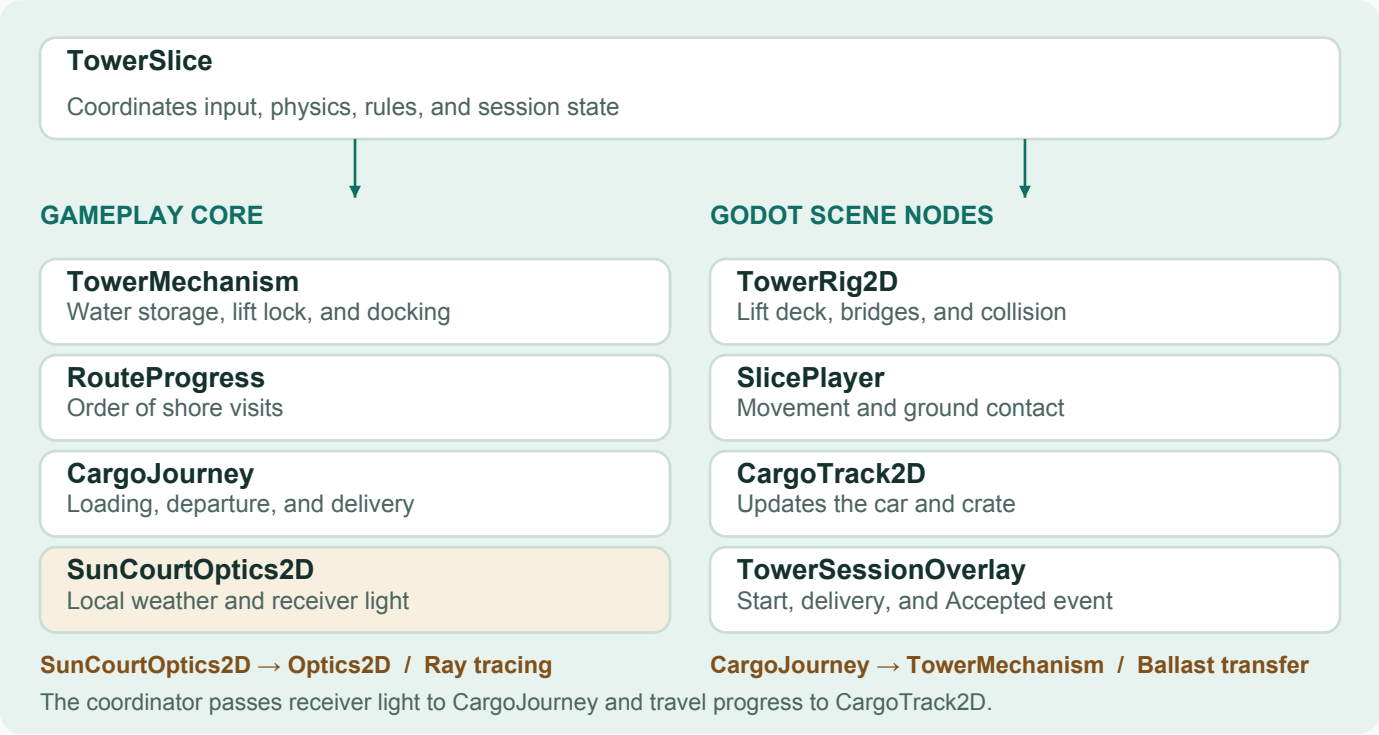
05B · 3D east receiver lit

Figure 1. East receiver in the original 2D (Round 02) and 3D (Round 05B) game captures.

2D Class Relationships

03 · Frozen Baseline 6175db9

TowerSlice receives Godot input and coordinates the core rules, scene nodes, and session UI each physics frame. The rules own gameplay state for water, visits, light, and cargo. The player and scene adapters handle footing, collisions, rendering, and reset wiring.



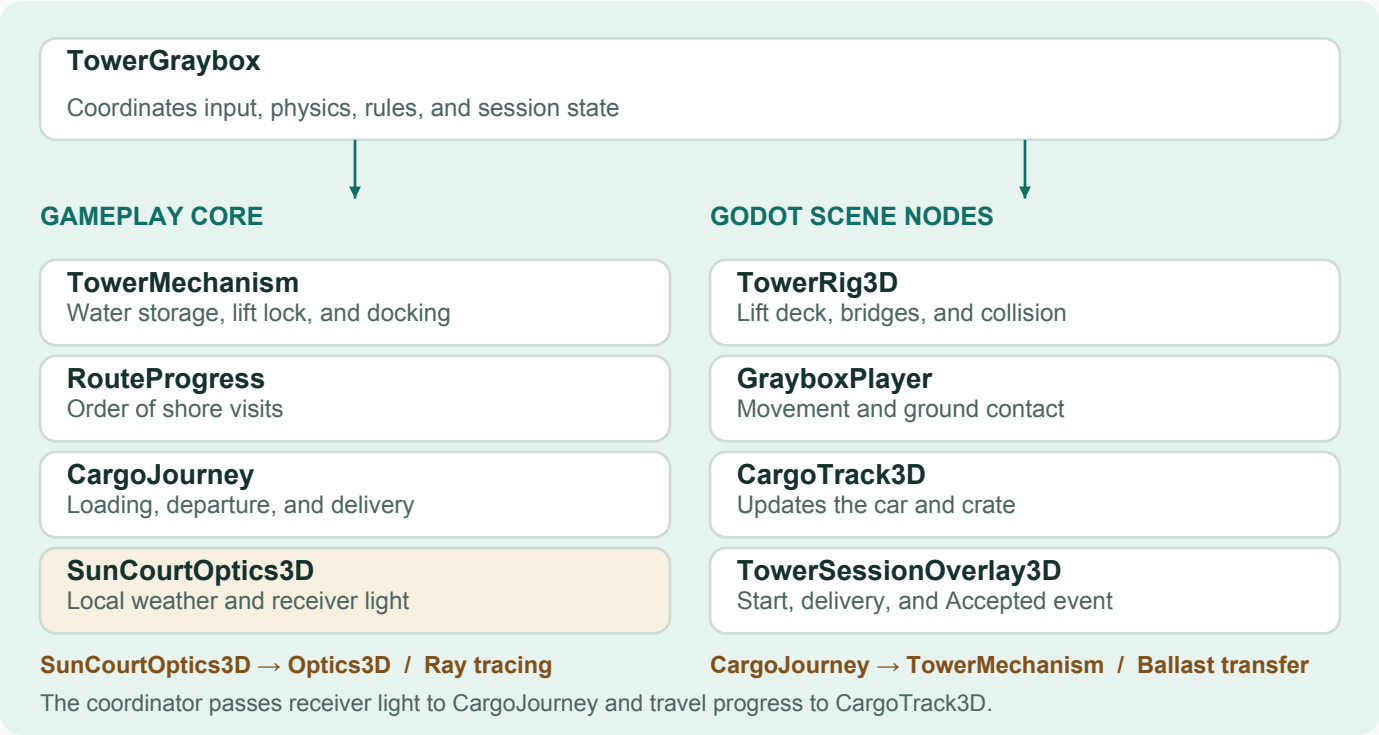
Class and responsibility	Key state, method, or event
TowerMechanism · Tower and rainwater	Exposes TowerState to synchronize the scene. Owns height, locking, docking, and water transfer.
RouteProgress · North before east	Visit(Port) and Reset(). Shore milestones and final delivery are separate.
CargoJourney · State of one fruit crate	Uses live receiver light to load cargo, transfer water, and authorize departure. Exposes travel progress and delivery count.
SunCourtOptics2D · Local weather and calibration	Owns the collector mirror's state and the light path. Delegates calculation to Optics2D.
TowerRig2D, SlicePlayer, CargoTrack2D · Godot adapters	Synchronize the tower, bridge, player footing, and physical cargo car.
TowerView, TowerSessionOverlay · Presentation	Read scene state. The opening and delivery cards relay input through Accepted.

Figure 2. Selected 2D relationships on page 3 of the PDF. Solid lines show coordinator ownership or calls. Gold arrows show dependencies for geometry and water transfer. Full 2D relationship map · 2D class map notes

3D Class Relationships

04 · Round 05D Candidate fb4cbb4

TowerGraybox replaces the 2D coordinator and links the same tower, route, and cargo sources. The physical tower's yaw sets the mirror pose. The optics adapter queries occlusion and sends the receiver result to the coordinator. The shared journey state still drives the car, with its position and collisions handled by the 3D scene.



Class and responsibility	Key state, method, or event
TowerMechanism, RouteProgress, CargoJourney · Shared rules	Own tower water and docking, visits, and cargo state respectively. The 3D project links all three source files directly.
SunCourtOptics3D · Local weather and calibration	Reads the tower mirror's world pose, raycasts for occlusion, and exposes TowerHit and ReceiverLit.
Optics3D · Pure geometry solver	Uses finite discs and world-space vectors to compute intersections and reflections, returning LightPath3D.
TowerRig3D, GrayboxPlayer, CargoTrack3D · Godot adapters	Synchronize tower yaw, bridge collision, player movement, and the physical car.
TowerSessionOverlay3D · Session UI	Shows the opening or delivery screen according to TowerScreen. Accepted returns control to the existing restart path.
GrayboxRouteCheck · Verification entry point	Sends key events and checks footing, light, cargo, delivery, and reset.

Figure 3. Selected 3D relationships on page 4 of the PDF. The geometry and scene wiring use 3D implementations. The resulting light path drives the same cargo rules. Full 3D relationship map · Compile links

Change Map and CCR

05 · Core Change Ratio for the Round 05D Candidate

Frozen 2D core class	3D counterpart	Change and rationale
TowerMechanism	Shared source	Retains water storage, lift, mechanical lock, heading, docking, and transfer to ballast.
RouteProgress	Shared source	Preserves the north-then-east order. Each scene supplies the physical visit locations.
CargoJourney	Shared source	Retains light-gated loading, water transfer, departure, travel, and delivery confirmation.
Optics2D	Optics3D	Replaces finite segments with finite discs and world-space intersections. Retains two calculated reflections and occlusion.
SunCourtOptics2D	SunCourtOptics3D	Retains weather and calibration. Uses the physical tower mirror's pose and Physics3D for occlusion.

Candidate CCR = (2 modified + 0 removed) ÷ 5 frozen core classes × 100% = 40.00%. Three classes share source, and 3D adds zero independent core responsibilities. Each optics replacement counts as one modification to an existing responsibility.

Scope. The frozen baseline fixes the denominator at five core classes. The optics controller counts because it owns weather, mirror, and receiver states that affect gameplay. Input mapping, character physics, scene adapters, camera, UI, scene assets, and tests are tracked separately as porting work.

Type of change	2D → 3D
Coordination and controls	TowerSlice → TowerGraybox. Side-view movement becomes planar movement. W/S crosses the physical north bridge.
Scene and presentation	TowerRig2D and CargoTrack2D gain 3D equivalents. The side-view section becomes a fixed-view 3D graybox. The opening and delivery screens convey the same information.
Additions and removals	Added a 3D scene, input and physics adapters, and a verification entry point. All frozen core responsibilities are retained. The 2D project still runs independently.

Reproduce the calculation. From the repository root, run python3 Tools/ccr.py --analysis BorrowingSun3D/Docs/ccr-analysis.json. The tool verifies the tag, commits, Git source objects, and candidate mappings. We classify behavior changes by reviewing each class. The 40% measures implementation changes in these five classes. Total effort and gameplay equivalence require separate evidence. If the final candidate's gameplay code changes, review the calculation against that code. Class-by-class change map · Frozen manifest · CCR log

Implementation and Acceptance

06 · Reproducible Delivery Route

Both versions use Godot 4.7.1 .NET and the .NET 8 SDK. Open either project's project.godot and press F5. Keep the projects side by side so 3D can link 2D source. The tests send key events to the live scene and use a temporary beam blocker. Replay uses the default scene. Run instructions · Shared acceptance cases

Environment and method	Frozen 2D 6175db9	3D candidate fb4cbb4
macOS 26.5.2 arm64	154 headless, 46 windowed. One normal delivery in the windowed run.	160 headless, 160 windowed. Includes refused departures, two deliveries, and replay.
Hosted Windows Server 2025 x64, headless	154 passed.	160 passed.
Hosted Windows Server 2025 x64, windowed	90 passed. Two deliveries and a low bridge crossing after R.	160 passed. Two deliveries. Tests refusal for low ballast and loss of light, plus travel after loss of light.
Manual Windows PC acceptance	Pending. Check the default driver, focus, mouse input, and two deliveries.	Pending on the same candidate. Capture screenshots and logs.

Reproducible scenario	Observed behavior
Low ballast or loss of light before departure	Pressing G leaves the car at the dock, with transferred water still in the ballast. The 2D headless and 3D routes test both cases. The hosted 2D windowed run records the waiting prompt.
Full ballast and live light	Press G aboard the car to cross the valley. Press G at the Flower Court marker to record 1/1.
Deliver again after R	The tower, mirrors, ballast, and count reset. Cross the low bridge from West Bank and repeat the full route.

In the hosted 3D windowed test, the car carried the grounded player 7.24 m and arrived after losing light on the first trip. Total water and physical tower height showed zero drift across 1,607 physics samples. Raw route log



2D · macOS delivery capture



3D · macOS delivery capture

Figure 4. Both macOS delivery screens show 1/1. Assertion counts reflect each probe's coverage. macOS logs · Windows results and rerun commands

Screenshots from Hosted Windows Tests

07 · Left 2D Baseline · Right 3D Round 05D Candidate

The captures were taken on 2026-11-09 Singapore time in a GitHub Actions Windows Server 2025 graphical session. The Windows display driver rendered through ANGLE and Direct3D 11 at a 1028×578 viewport. Both routes passed on the first run. Pixel rounding caused a false failure in the screenshot aspect-ratio check. After the QA fix, the full rerun passed. These original captures are from the first run. First-run artifact · Passing rerun



Opening screen · 2D



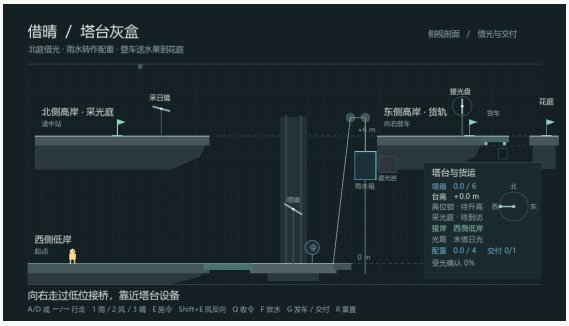
Opening screen · 3D



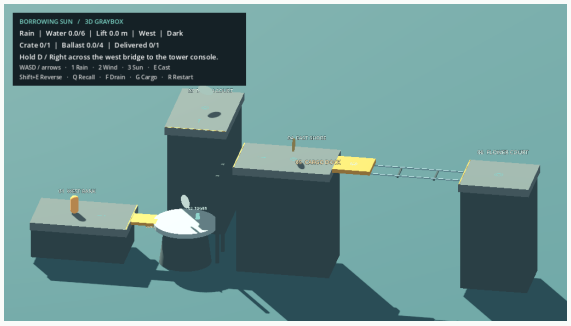
First delivery 1/1 · 2D



First delivery 1/1 · 3D



R resets to the west low bridge · 2D



R resets to the west low bridge · 3D



Second delivery 1/1 · 2D



Second delivery 1/1 · 3D

Figure 5. Four checkpoints in play order. The route logs record the low bridge crossing and both deliveries. Manual Windows PC results are pending. 2D log · 3D log · Full capture index

Development Evidence and Submission Tasks

08 · Shared Group Draft

Evidence	Records to check
Version history	2d-baseline tag · 05D candidate · GitHub comparison from 2D to 3D
Main development	buriallsy/3d-graybox · Start of the 3D port in 05A · Acceptance evidence commit
Team collaboration	Git author djwcb2333 contributed the domain architecture and numeric boundaries. Merge commit 6b4357a retained Round 03 gameplay and integrated standalone domain tests, CI, CCR tooling, and rerun notes. Members will confirm their own names, branches, contributions, and assessments.
AI collaboration index	An existing member index documents work and checks on scope, values, and the framework. The Windows preflight is another example. With a physical Windows PC unavailable, the user chose hosted windowed testing. The first-run logs, QA fix, and passing rerun document how the AI-generated check was verified. Members record their full exchanges and which suggestions they used. Existing member index · QA fix

Individual signed reflections, pending. Each member should give their name and branch, describe a specific decision or implementation they owned, and explain the problem, how they verified the result, and how course principles shaped their work. Include how they checked and revised AI output, with a commit or pull request and a private transcript index. The group must add its number, official member list, individual reflections, and signed declarations.

Next acceptance run. Follow the manual Windows checklist on a Windows PC for both opening screens, two deliveries, and the low bridge crossing after reset. Record the operator, environment, source SHA, and screenshots. Record first-time players' completion times and sticking points separately. After acceptance on both platforms, confirm the final code against the candidate before deciding on 3d-final and the CCR.

Submission. Rehearse the 15-minute demo on the same code versions. The final archive should contain both runnable projects, a README, the joint PDF, the group originality declaration, and each member's generative AI declaration. A member should test launch from a freshly extracted copy before submission. Check the course portal for the group number, deadline, naming rules, and declaration templates. This draft follows the repository's summary of the course requirements. Check the original brief before submitting. Production plan · Packaging checklist

Owner before finalization	Evidence to supply
Windows test operator	PC logs for both versions, opening and delivery captures, low bridge replay, and reviewer.
Each member	Name, evidence of their contribution, individual reflection, and full AI transcript index or a declaration of non-use.
Packaging and submission owner	Confirm the group number, deadline, signatures, and filenames. Verify launch after extraction and save the submission receipt.